# Automatic Car parking system

**A PROJECT REPORT**

*Submitted by:*


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*In fulfilment for the award of the degree*

*of*

**BACHELOR OF ENGINEERING**

*in*

**Electronics and Communication**


**LDRP Institute of Technology and Research, Gandhinagar**

**Kadi Sarva Vishwavidyalaya, Gandhinagar**

## *LDRP INSTITUTE OF TECHNOLOGY AND RESEARCH, GANDHINAGAR*

## Electronics and Communication Engineering Department

# CERTIFICATE

This is to certify that the Project Work entitled **"Automatic Car parking system"** has been carried out by **Meet Rudani ( 20BEEC30006 ) ,Parth Vaghasiya ( 20BEEC30007)** under my guidance in fulfilment of the degree of Bachelor of Engineering in Electronics and Communication (7th Semester) of Kadi Sarva Vishwavidyalaya University, Gandhinagar during the academic year 2023.

| | |
|---|---|
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| Name & Sign | Name & Sign |
| **Internal Guide** | **Head of the department** |
| **LDRP-ITR** | **LDRP-ITR** |

# ACKNOWLEDGEMENT

The project report on " **Automatic Car parking system**" was taken up as 7th Semester Project. Here I wish to express my sincere thankfulness to those people who have helped me throughout my project work. I would like to thank the people who stood by me to prepare this-project report. I sincerely feel that the credit of the project work cannot be narrowed down to only one individual.

This project work is an integrated effort of all those person concerned; through co-operative and effective guidance I would achieve its completion. I would like to thank **Dr. Kruti P. Thakore**, Prof. EC Department, for making available all facilities in fulfilling the requirement for project work. I also take this opportunity to express a deep sense of gratitude to **Dr. SHRIDHAR MENDHE**, HOD EC Department for his cordial support, valuable information and guidance, which helped us in completing this task through various stages.

I am deeply appreciative of our parents and our colleagues for their constant support, encouragement and bearing with us during the period of our project work. Finally, it is our foremost duty to thank all our respondents, who helped us to complete our Project, without them this project was not possible.

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# Abstract

In current times, the concept of smart cities have gained much popularity. Consistent efforts are being made to maximize the productivity and reliability of urban infrastructures. Parking is costly and limited in almost every major city of India. Innovative parking systems for meeting near term parking demand are needed. Due to rapid increase in vehicle density especially during the peak hours of the day, it is a tedious task for the public to find a secured parking space for their vehicles. This proposes a novel, secured, efficient and intelligent parking system based on Arduino and sensors. The proposed smart parking system includes of an onsite slot module development that will monitor the availability of each parking slot. Due to number of vehicles on road, causes traffic problems our smart parking system helps us to find parking spaces in parking lot with the implementation of smart parking system patrons can easily locate vacant parking spaces subsequently , the various sensor systems used in developing the systems and our car parking system reducing congestion in cities and our proposed models such that no man is needed to park the cars and no receipt system problem in our project has the automatic receipt system and this project helps in reducing the average time of users for that they wait for the parking of car all such systems will coordinate they do not require to wait we have automatic door opening and closing system using Servo motor by using push buttons such that car enter and exit though entry and exit parking system we have created this project to 7 cars we can increase the parking spaces such that many cars are parked in our parking lot and we have used IR sensors for visualization . Further modifications includes a mobile application that allows the user to track the availability of parking space in the required locality. This smart parking can increase the economy by lowering the labour cost, reducing the fuel consumption and pollution in urban cities.

# CHAPTER -1

## INTRODUCTION

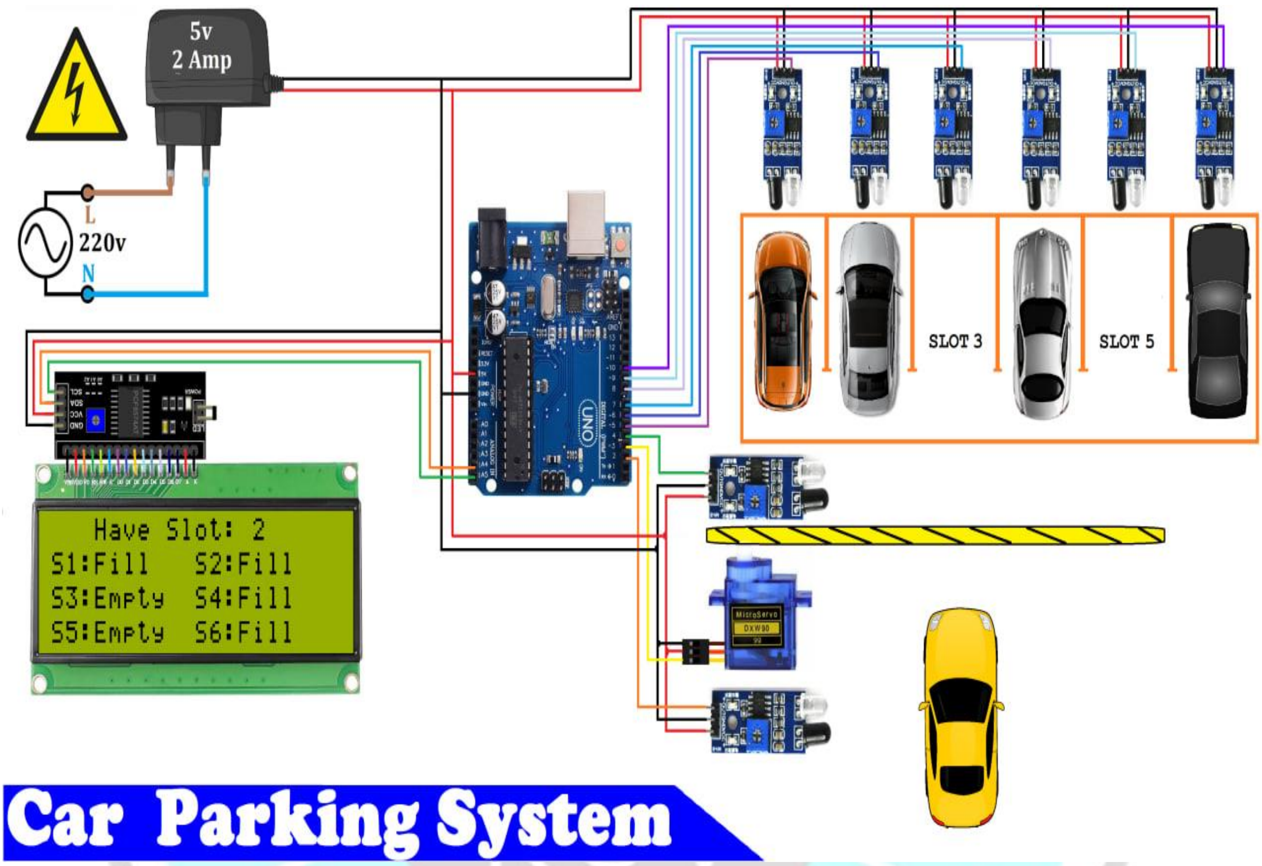
Whenever we plan for an outing, the major concern that troubles us is the availability of parking space at the desired location. A lot of fuel is wasted in searching proper parking space, which increases the pollution as well as cost. Generally, the parking areas are managed by some private agencies that appoint people to help in finding parking space and they charge an amount which could be minimised if we opt for smart parking system. This was the problem case for our minor project. In the development of traffic management systems, an intelligent parking system was created to reduce the cost of hiring people and for optimal use of resources for car - park owners. Currently, the common method of finding a parking space is manual where the driver usually finds a space in the street through luck and experience. This process takes time and effort and may lead to the worst case of failing to find any parking space if the driver is driving in a city with high vehicle density. The alternative is to find a predefined car park with high capacity. However, this is not an optimal solution because the car park could usually be far away from the user destination. In recent years, research has used vehicle-to-vehicle and vehicle-to infrastructure interaction with the support of various wireless network technologies. However, the current intelligent parking system does not provide an overall optimal solution in finding an available parking space, does not solve the problem of load balancing, does not provide economic benefit, and does not plan for vehicle-refusal service. To resolve the afore mentioned problems and take advantage of the significant development in technology, the Arduino based applications has created a revolution in many fields in life as well as in smart-parking system (SPS) technology. This research also implements a system prototype with wireless access in an opensource physical computing platform based on Arduino technology using a smartphone that provides and the vehicles to verify the feasibility of the proposed system.

## Merits:

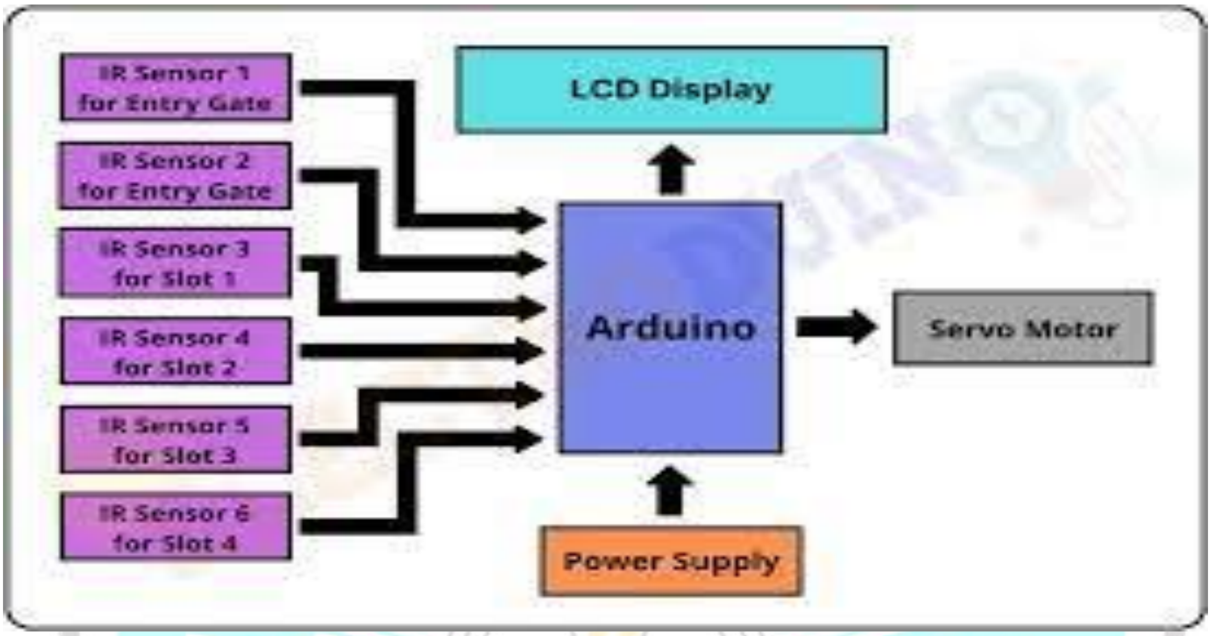
In order to make smart parking system we have to come up with a robust system that should be efficient as well as uses least resources. :

- **Optimized parking** – Users find the best spot available, saving time,resources and effort. The parking lot fills up efficiently and space can be utilized properly by commercial and corporate entities.
- **Reduced pollution** – Searching for parking burns around one million barrels of oil a day. An optimal parking solution will significantly decrease driving time, thus lowering the amount of daily vehicle emissions and ultimately reducing the global environmental footprint.
- **Real-Time Data and Trend Insight** – Over time, a smart parking solution can produce data that uncovers correlations and trends of users and lots. These trends can prove to be invaluable to lot owners as to how to make adjustments and improvements to drivers.
- **Decreased Management Costs** – More automation and less manual activity saves on labor cost and resource exhaustion.

# BLOCK DIAGRAM



- **PRODUCT FLOWCHART**

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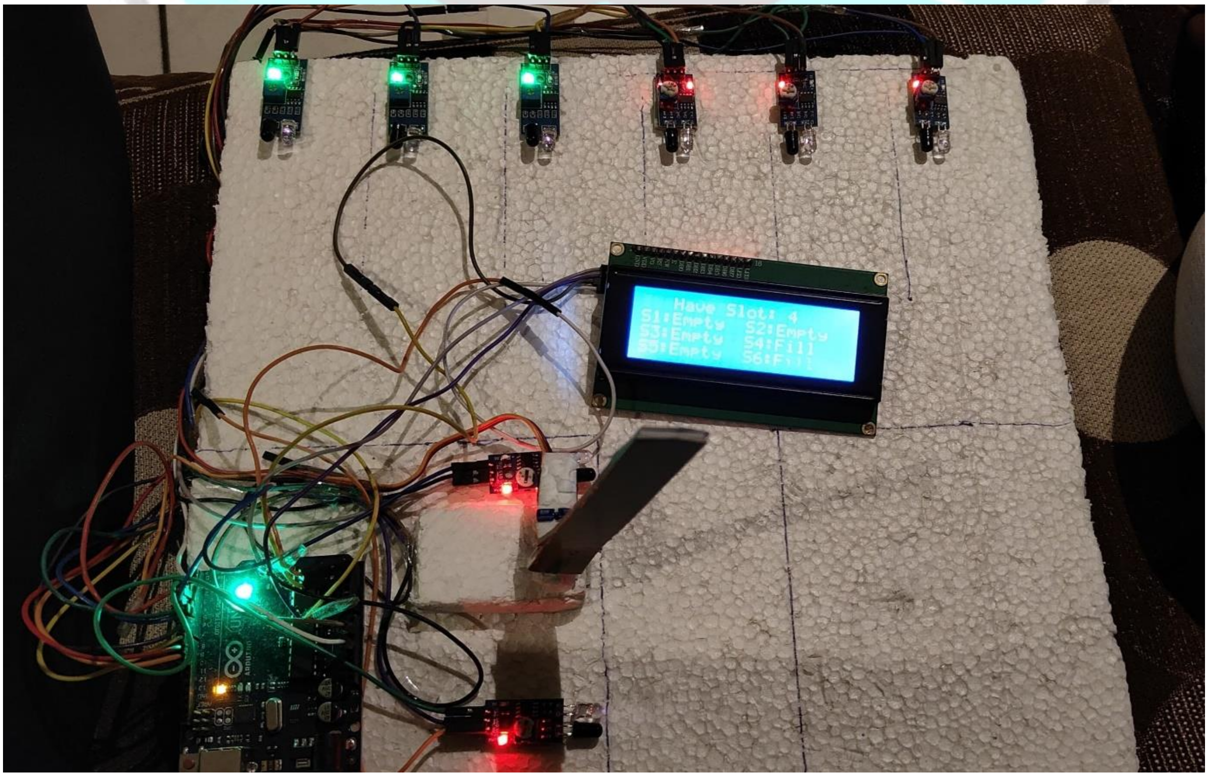


## Background or Problem

Making a parking monitoring system especially at Saudi Arabia will be significantly useful due to the increasing usage of parking lot as population and buildings increase more and more which indicates that there is really lack of parking and a large amount of traffic that look for free parking spaces. However, this would be time consuming for people to keep searching and Page 11 of 148 also there is an environmental issue since the automobiles is emitting CO2 which is a major factor in affecting environment. In order to deal with the problems of the metropolitan areas, there is a need to develop advanced solutions for efficient parking space management especially at airports, malls, universities, and any large parking lots that are used heavily. There are a number of companies that provide parking management solutions such as Siemens, Nortech and Omnitech.a brief introduction about each is described below but Omnitech was focused on due to the available information they provided regarding used technology.

Siemens provides several parking management services including: intelligent parking guidance systems, convenient pay-and-display machines, and modern car park technology. It uses Sitraffic Guide: parking guidance system which uses dynamic message signs to provide information to the driver on the current occupancy levels of individual parking areas and guide people along the shortest route to final parking destination which will dramatically reduce traffic volumes and environmental pollution.

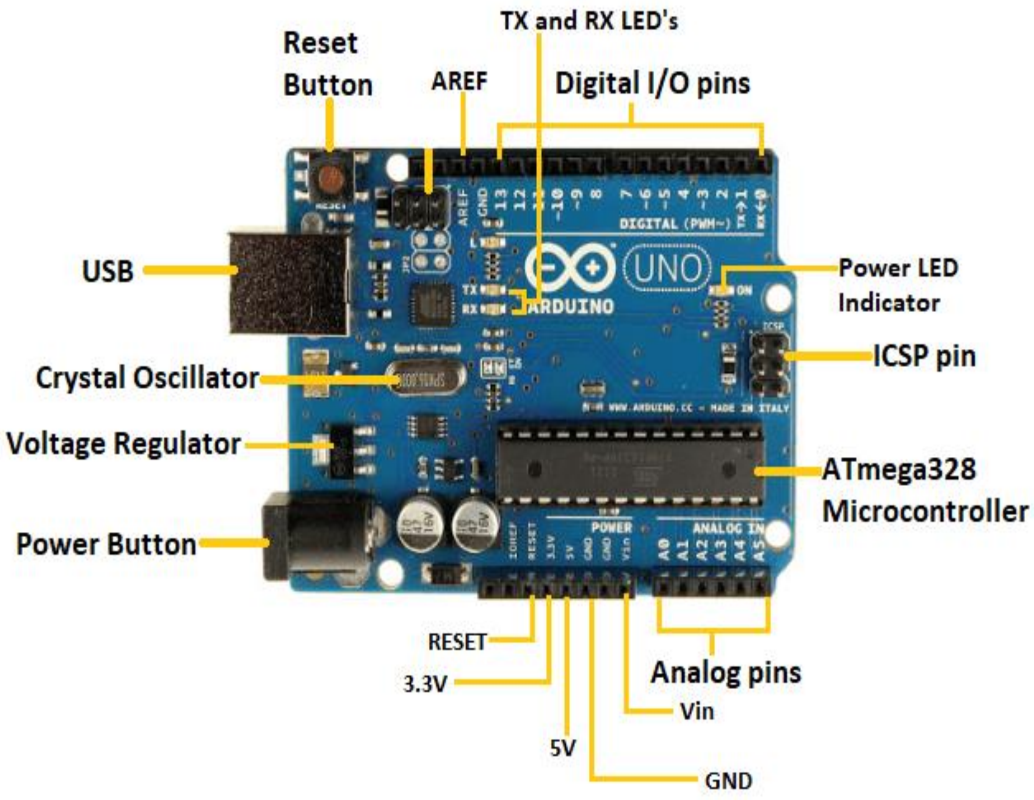
Nortech uses group of products that provides methods of counting vehicles entering or leaving a car park or depot and keeping track of available spaces. it uses what they call it NorParc Level Counting and Guidance System which offers real-time on-screen information feedback, alarms and report generation. It can manage both small, single level car parks and large multi-storey car parks with signs showing available space on each level.

Omnitech uses modern parking management systems which consists of entrance and exit gate barrier units, parking guidance, car parking control and garage payment system, and parking counting systems in all various types which are all connected from manual to automatic pay stations and a car park management center. The main services that are relevant and related to this context is parking guidance system and parking counting system.One type of sensor is the OmnyPark OP-511 which is an ultrasonic parking sensor that is used for garage parking guidance system designed with an ultrasonic detector use as one of the main part of the parking guidance and management system solution. Parking sensor makes use of ultrasonic detecting the vehicle parking space to get message and transfer to Zone Control Unit or other equipment through the RS485 connection. One parking space needs one Ultrasonic Detector which is installed in the middle of the car park space.

# CHAPTER -2

## Components Used

1. **Arduino:** In this project we have used arduinoUNO board . It is microcontroller board based on the ATmega328P. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz quartz crystal, a USB connection, a power jack, an ICSP header and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started. The Arduino/Genuino Uno can be programmed with the (Arduino Software (IDE)). The ATmega328 on the Arduino/Genuino Uno comes preprogrammed with a bootloader that allows you to upload new code to it without the use of an external hardware programmer. The Arduino/Genuino Uno board can be powered via the USB connection or with an external power supply. The power source is selected automatically. The ATmega328 has 32 KB (with 0.5 KB occupied by the bootloader). It also has 2 KB of SRAM and 1 KB of EEPROM (which can be read and written with the EEPROM library).



- **Power USB**

Arduino board can be powered by using the USB cable from your computer. All you need to do is connect the USB cable to the USB connection.

- **Power Button**

Arduino boards can be powered directly from the AC mains power supply by connecting to it.

- **Voltage Regulator**

The function of the voltage regulator is to control the voltage given to the Arduino board and stabilize the DC voltages used by the processor and other elements.

- **Crystal Oscillator**

The crystal oscillator helps Arduino in dealing with time issues. How does Arduino calculate time? The answer is, by using the crystal oscillator.

- **Arduino Reset**

You can reset your Arduino board, i.e., start your program from the beginning. You can reset the UNO board in two ways. First, by using the reset button on the board. Second, you can connect an external reset button to the Arduino pin labelled RESET.

- **Pins(3.3, 5, GND, Vin)**

➢ 3.3V (6) − Supply 3.3 output volt

➢ 5V (7) − Supply 5 output volt

➢ Most of the components used with Arduino board works fine with 3.3 volt and 5 volt.

➢ GND (8)(Ground) − There are several GND pins on the Arduino, any of which can be used to ground your circuit.

➢ Vin (9) − This pin also can be used to power the Arduino board from an external power source, like AC mains power supply.

- **Analog pins**

The Arduino UNO board has six analog input pins A0 through A5. These pins can read the signal from an analog sensor like the humidity sensor or temperature.

## • ATmega 328 microcontroller

It is a single chip Microcontroller of the ATmel family. The processor code inside it is of 8-bit. It combines Memory (SRAM, EEPROM, and Flash), Analog to Digital Converter, SPI serial ports, I/O lines, registers, timer, external and internal interrupts, and oscillator.

## • ICSP pin

The In-Circuit Serial Programming pin allows the user to program using the firmware of the Arduino board.

## • Power LED indicator

This LED should light up when you plug your Arduino into a power source to indicate that your board is powered up correctly. If this light does not turn on, then there is something wrong with the connection.

## • TX and RX LEDs

On your board, you will find two labels: TX (transmit) and RX (receive). They appear in two places on the Arduino UNO board. First, at the digital pins 0 and 1, to indicate the pins responsible for serial communication. Second, the TX and RX led (13). The TX led flashes with different speed while sending the serial data. The speed of flashing depends on the baud rate used by the board. RX flashes during the receiving process.

## • Digital I/O

The Arduino UNO board has 14 digital I/O pins (15) (of which 6 provide PWM (Pulse Width Modulation) output. These pins can be configured to work as input digital pins to read logic values (0 or 1) or as digital output pins to drive different modules like LEDs, relays, etc. The pins labeled “~” can be used to generate PWM.

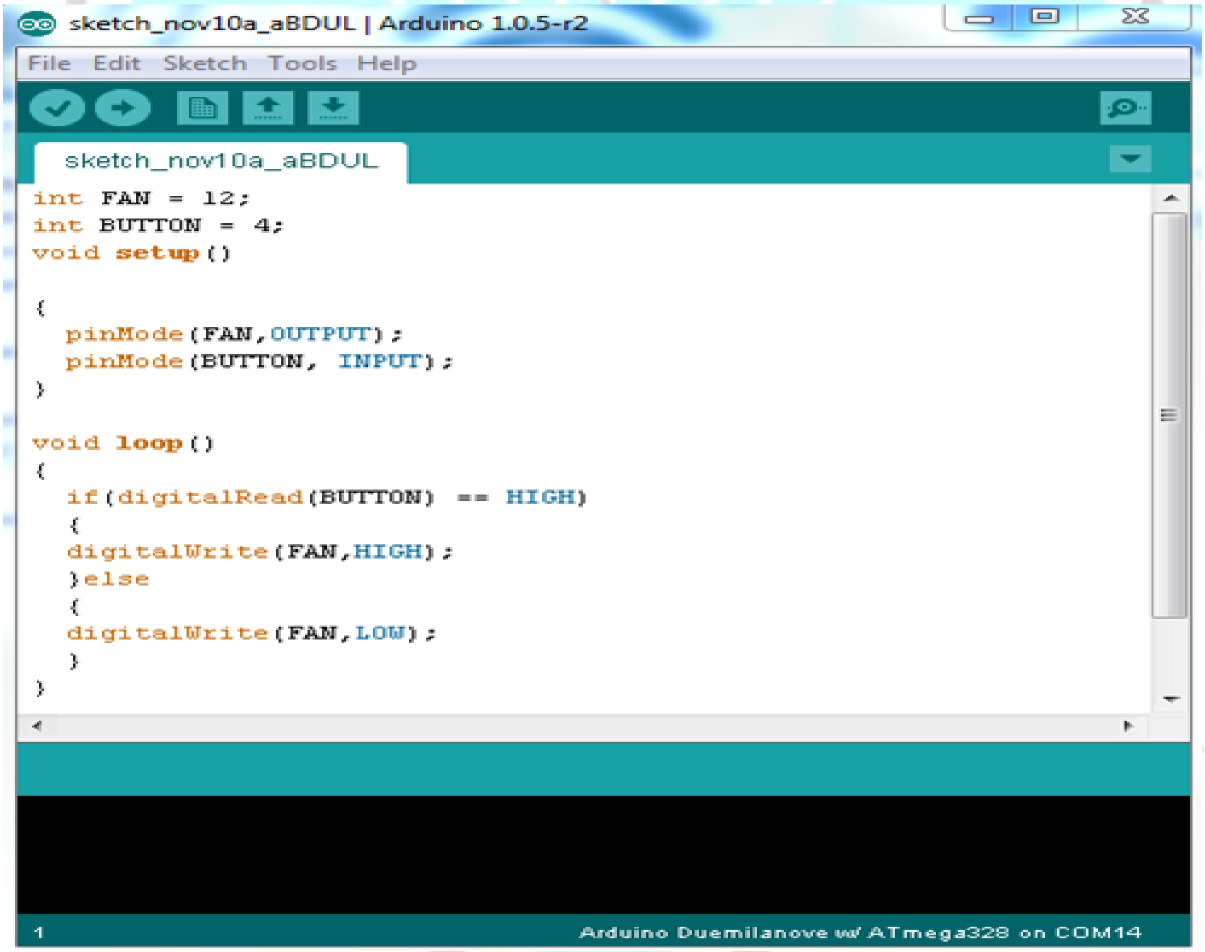
## • AREF

AREFstands for Analog Reference. It is sometimes, used to set an external reference voltage (between 0 and 5 Volts) as the upper limit for the analog input

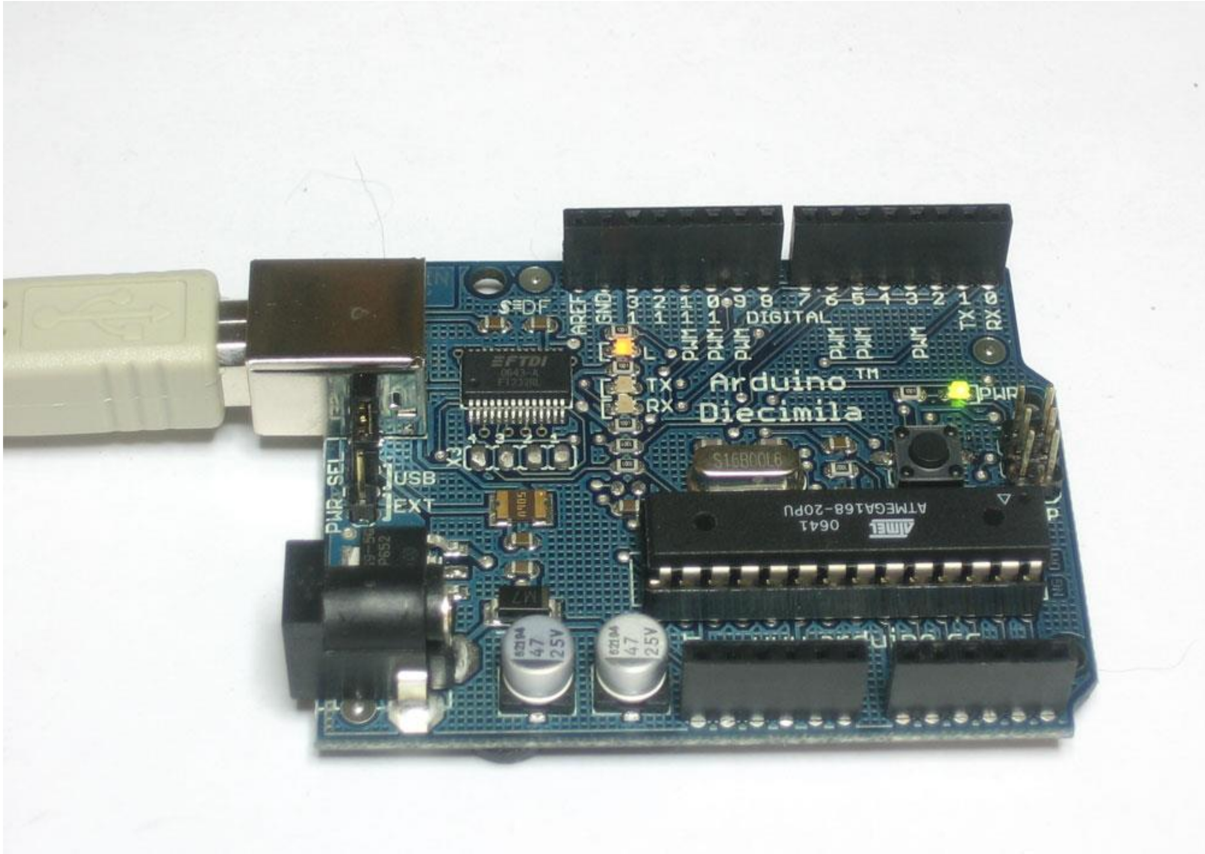
# Arduino Software, and Controller Chip

## IDE Software to program the Arduino

The Arduino has his own application software thatenables the programmer to download and upload programs and other functionalities such as debugging. Download the Arduino IDE (The software name) from the Arduino download page. This software uses C language as the programming language for theArduino as shown in figure.

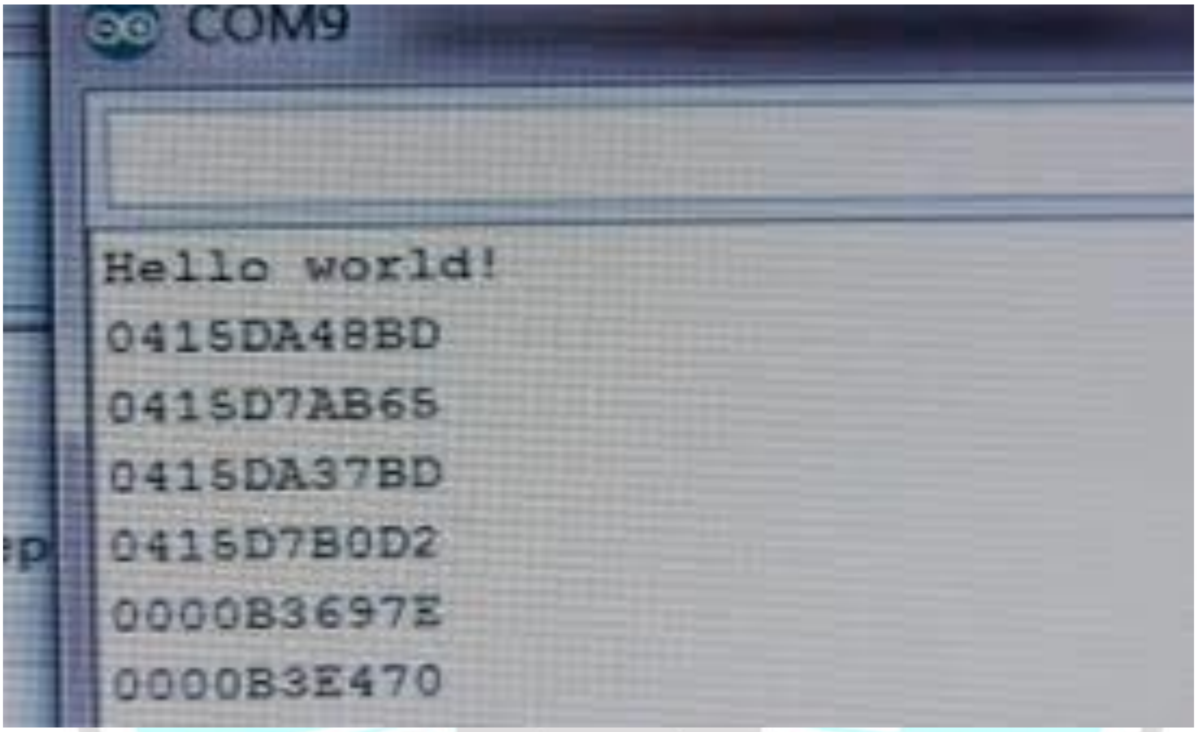



The communication between Arduino and PC is accomplished using USB cable to transfer data serially as shown figure .



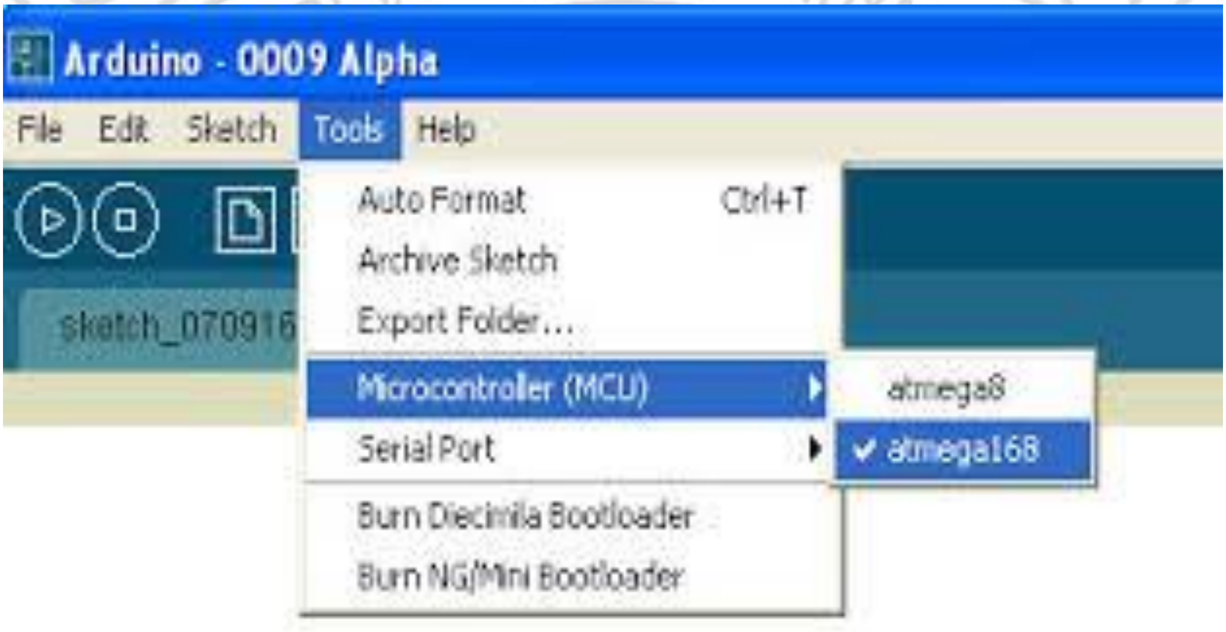

Seeing serial data requires a monitor, which like your display monitor will show us what data is being transferred as shown in figure .

We can also send serial data back to the Arduino using the Serial Monitor.The Baud Rate: default setting is 9600 baud for serial monitor as shown in figure.

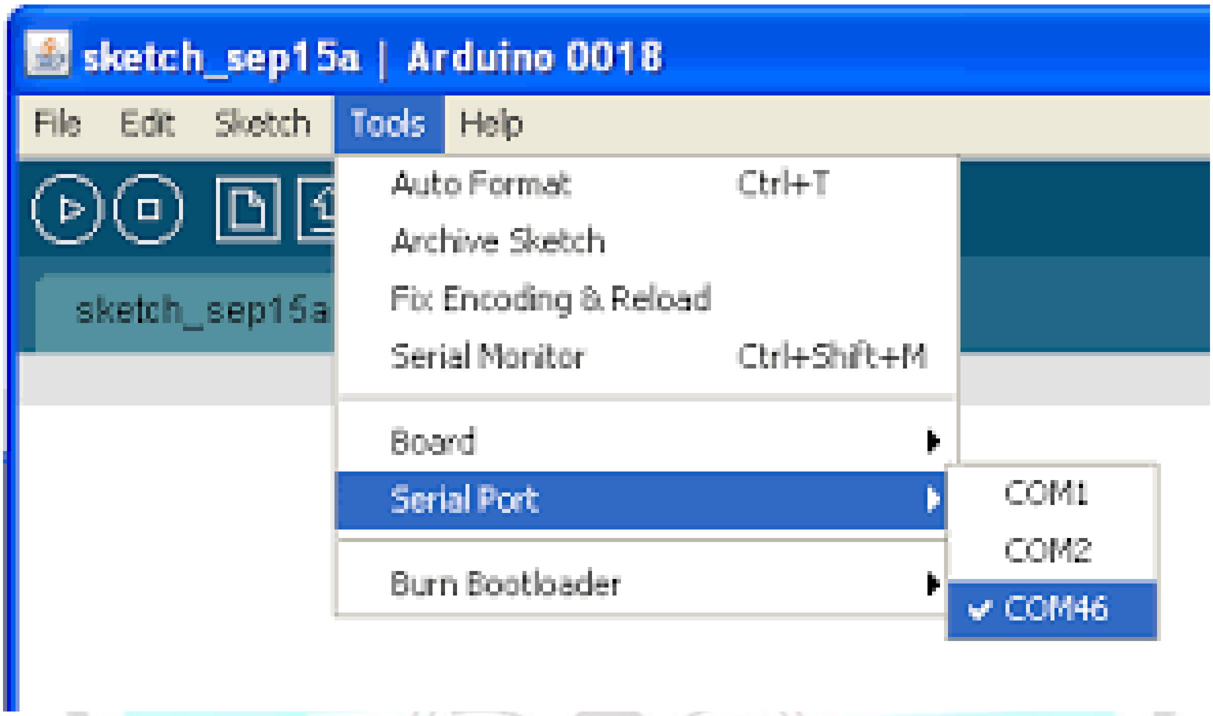



## Configuring the Arduino Software

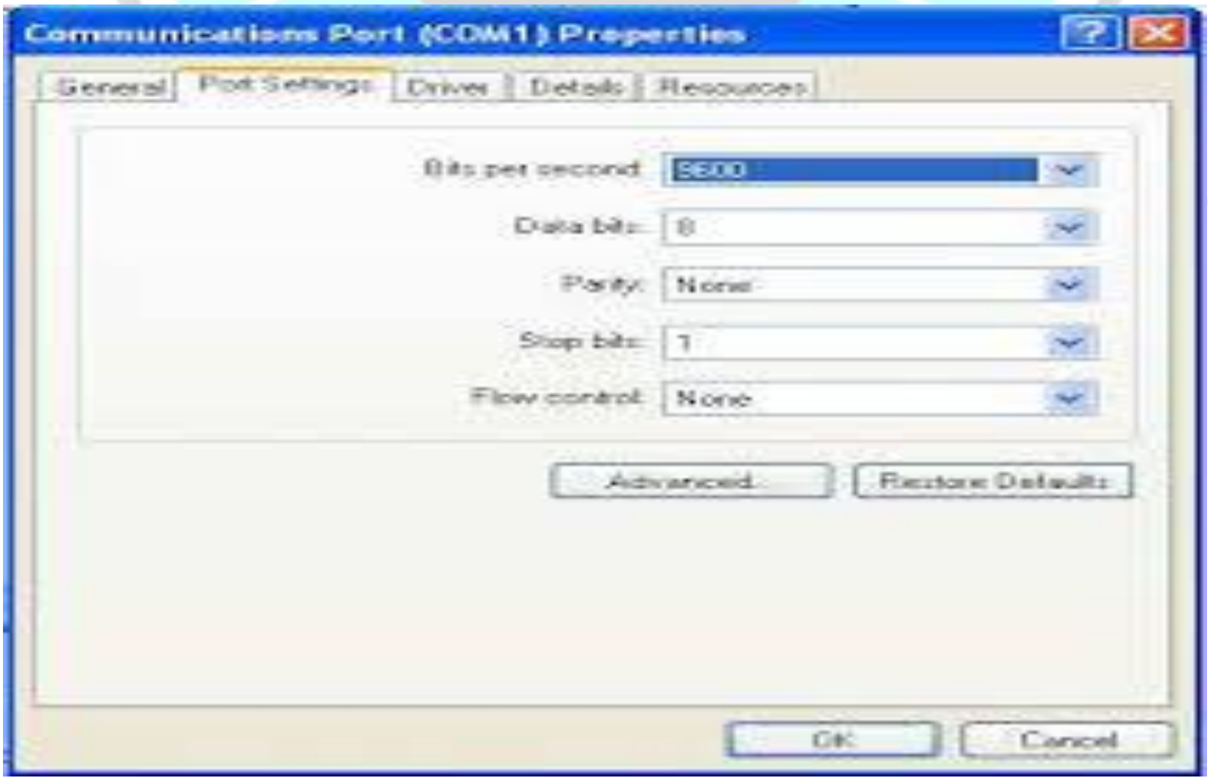
After downloading and installing the Arduino software, the first step is to configure the Arduino software for the correct chip. Look for the chip on the Arduino that looks like as shown in figure.



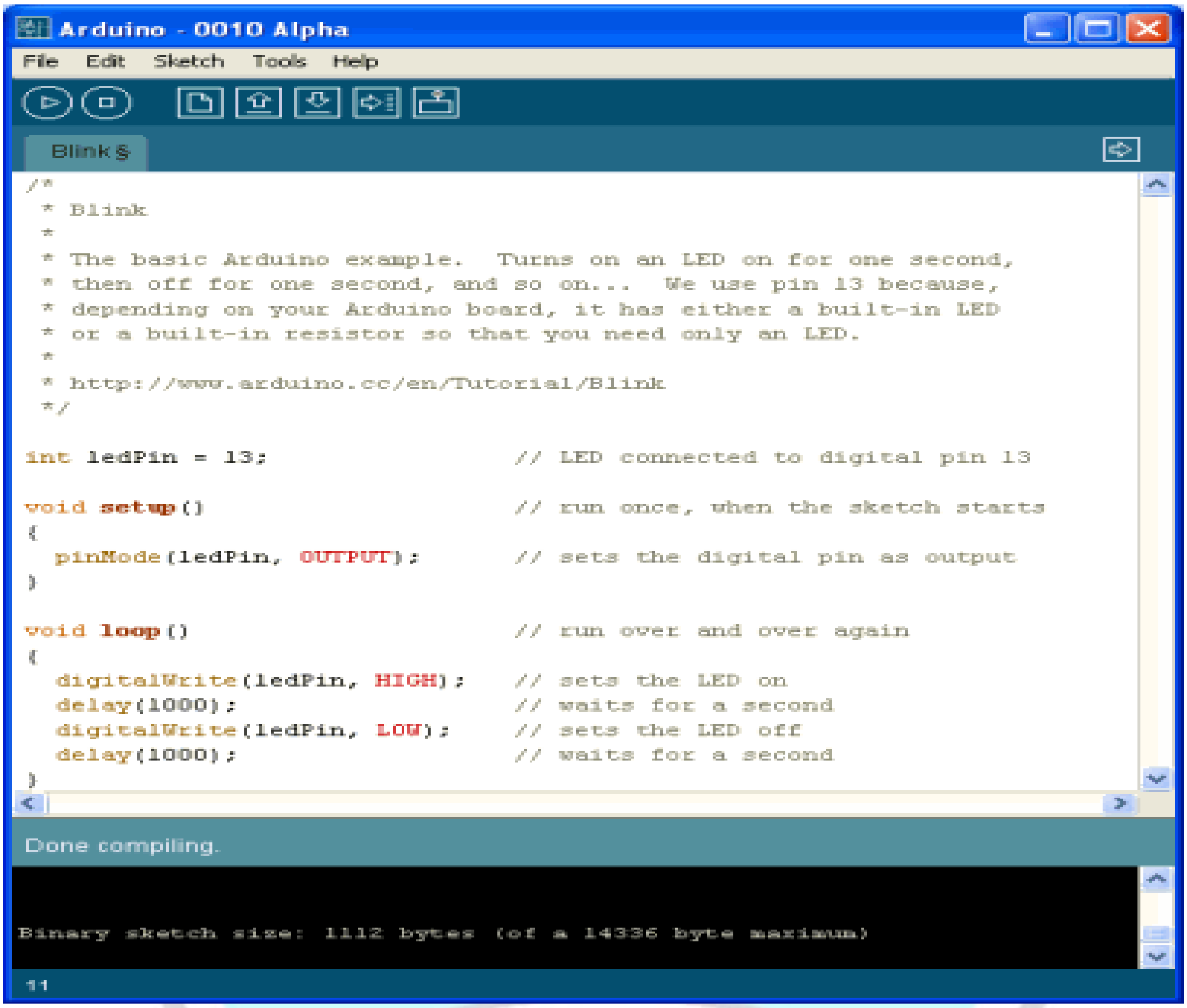
Next, is to configure the Serial Port: (See figure)




Set the PC communication baud rate to 9600 pits per second as shown figure



As shown in the figure below, the code can be written in this window:




After verifying the code, it can be sent to Arduino by pressing upload button

## Introduction about Coding of IDE Arduino Software

There are two main sections in this software. Below are some of main functions that are used in programming the Arduino and essential to learn for even simple programs:

**The setup ()** function is called when a sketch starts. It is used to initialize variables, pin modes, start using libraries, etc. The setup function will only run once, after each power up or reset of the Arduino board.

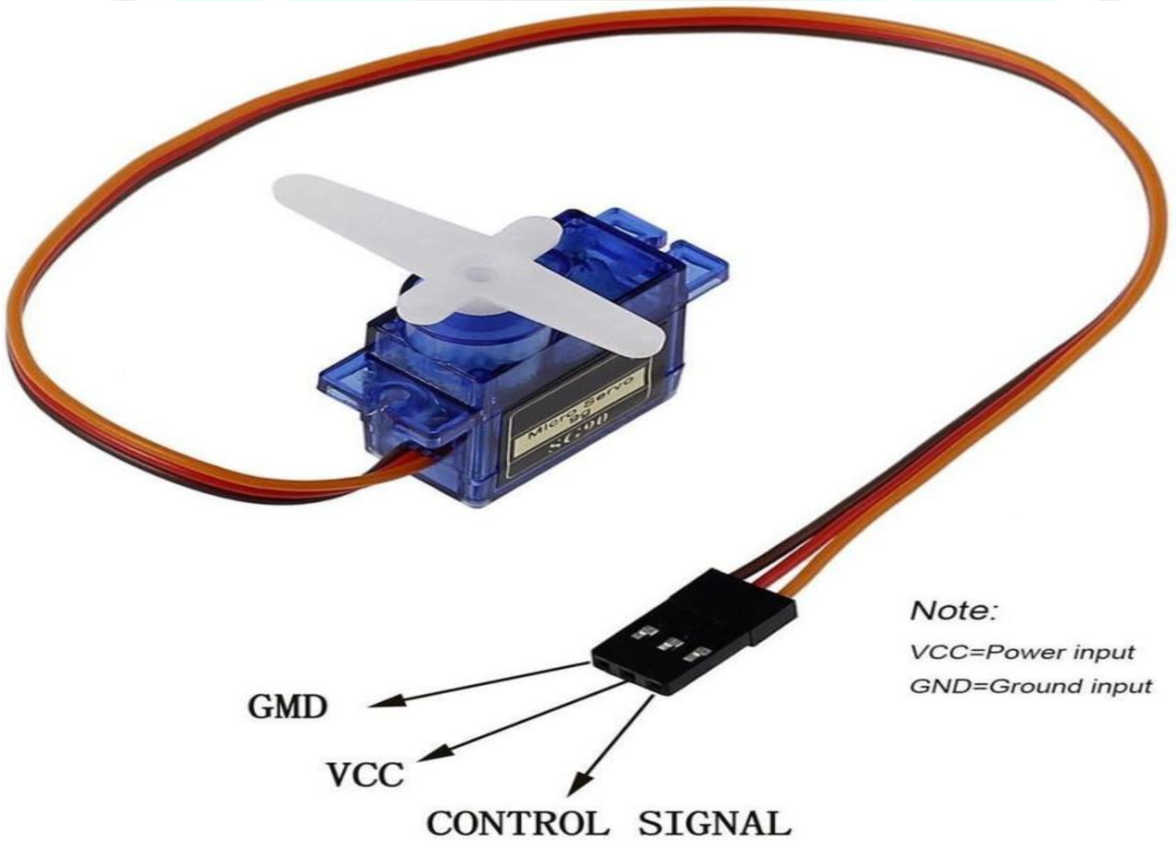
After creating a setup () function, which initializes and sets the initial values, **the loop ()** function does precisely what its name suggests, and loops consecutively, allowing your program to change and respond. It is used it to actively control the Arduino board.

#define is a useful C component that allows the programmer to give a name to a constant value before the program is compiled. Defined constants in Arduino do not take up any program memory space on the chip. The compiler will replace references to these constants with the defined value at compile time.

This can have some unwanted side effects though, if for example, a constant name that had been #defined is included in some other constant or variable name. In that case the text would be replaced by the #defined number (or text).

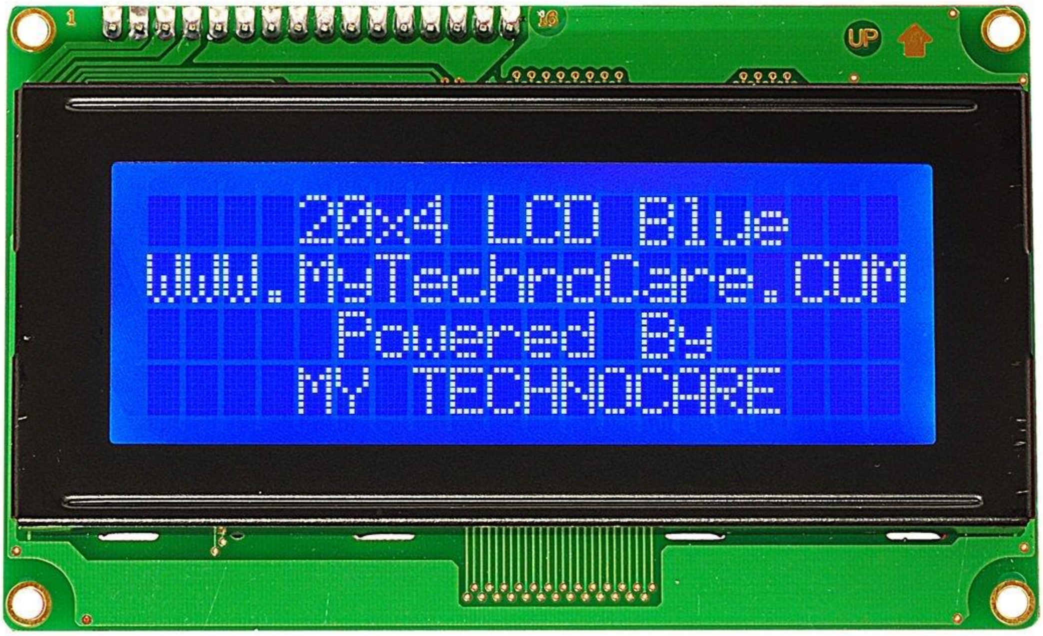
2.. **Servo Motor :**

A servomotor is a rotary actuator or linear actuator that allows for precise control of angular or linear position, velocity and acceleration. It consists of a suitable motor coupled to a sensor for position feedback. It also requires a relatively sophisticated controller (Arduino UNO in our case), often a dedicated module designed specifically for use with servomotors. A servomotor is a closed-loop servomechanism that uses position feedback to control its motion and final position. The input to its control is a signal (either analogue or digital) representing the position commanded for the output shaft. The motor is paired with some type of position encoder to provide position and speed feedback. In the simplest case, only the position is measured. The measured position of the output is compared to the command position, the external input to the controller. If the output position differs from that required, an error signal is generated which then causes the motor to rotate in either direction, as needed to bring the output shaft to the appropriate position. As the positions approach, the error signal reduces to zero and the motor stops.



3. **LCD Display :**

LCD (Liquid Crystal Display) is a type of flat panel display which uses liquid crystals in its primary form of operation. LEDs have a large and varying set of use cases for consumers and businesses, as they can be commonly found in smartphones, televisions, computer monitors and instrument panels.

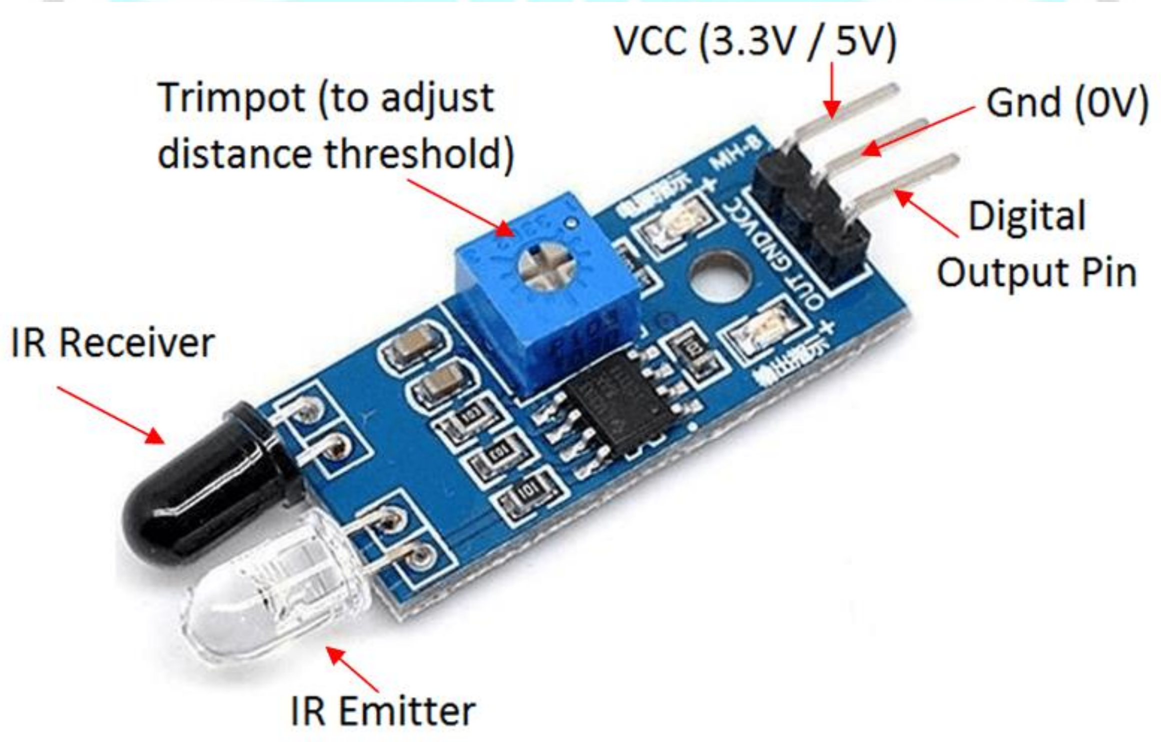



A display is made up of millions of pixels. The quality of a display commonly refers to the number of pixels; for example, a 4K display is made up of 3840 x2160 or 4096x2160 pixels. A pixel is made up of three subpixels; a red, blue and green—commonly called RGB. When the subpixels in a pixel change color combinations, a different color can be produced. With all the pixels on a display working together, the display can make millions of different colors. When the pixels are rapidly switched on and off, a picture is created.

We have used an LCD Display which shows the total number of slots and the free slots available in the parking area. If there is no parking spots available then it can show 'Space Full!"

4. **IR Sensors :**

IR sensor is an electronic device, that emits the light in order to sense some object of the surroundings. An IR sensor can measure the heat of an object as well as detects the motion. Usually, in the infrared spectrum, all the objects radiate some form of thermal radiation. These types of radiations are invisible to our eyes, but infrared sensor can detect these radiations.




The emitter is simply an IR LED (Light Emitting Diode) and the detector is simply an IR photodiode . Photodiode is sensitive to IR light of the same wavelength which is emitted by the IR LED. When IR light falls on the photodiode, the resistances and the output voltages will change in proportion to the magnitude of the IR light received.

There are five basic elements used in a typical infrared detection system: an infrared source, a transmission medium, optical component, infrared detectors or receivers and signal processing. Infrared lasers and Infrared LED's of specific wavelength used as infrared sources.

- **Elimination of the effect of the stray IR light**

To eliminate the effect of the stray IR light, the IR emitters are modulated and the output of the IR Detectors is demodulated to look for specific range of detection.

- **Features of Sharp IR Sensor**

• Measures distance in range from 10-80cm.

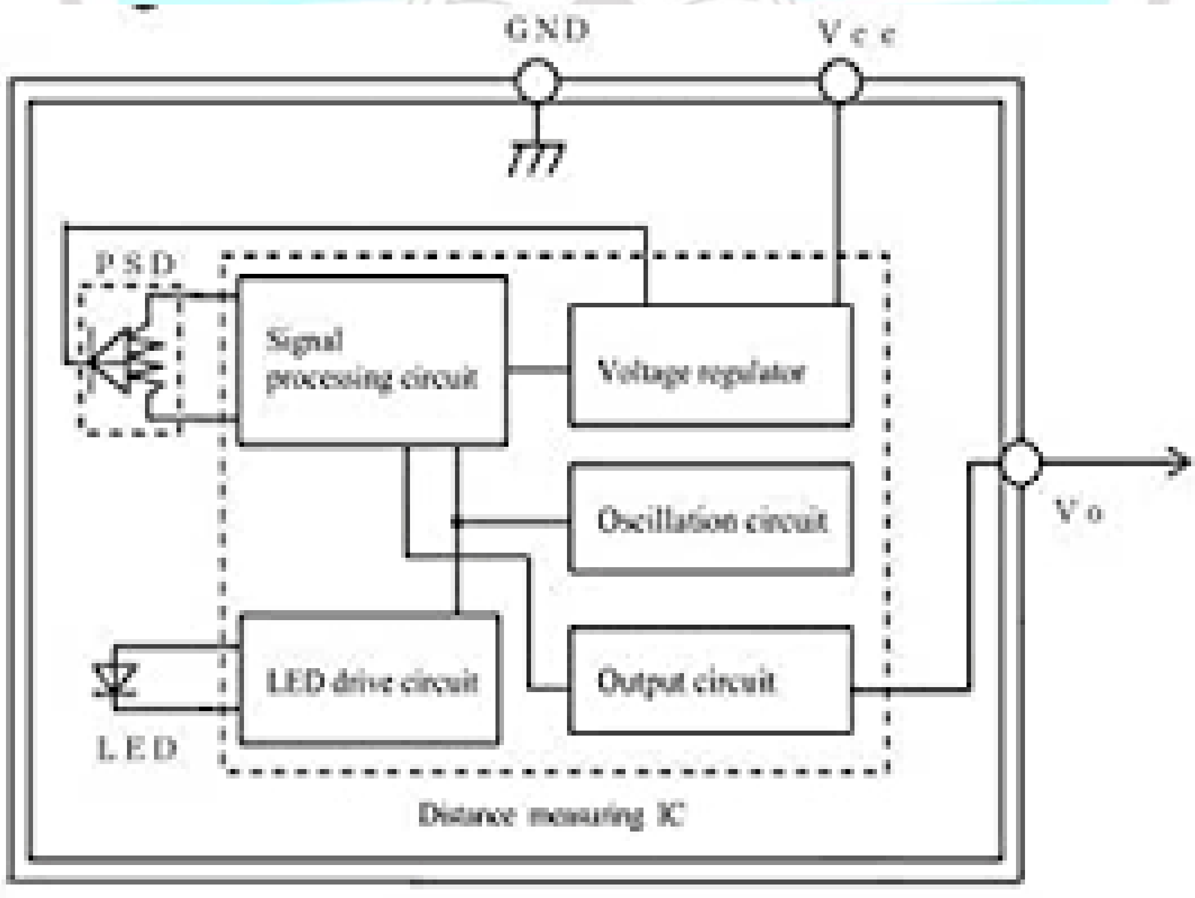
• Designed to interface to small micro-controllers.

• Less influence on the colors of the reflected objects and their reflectivity.

• Typical current consumption 30 mA.

• 3.1V at 10cm to 0.4V at 80cm.

• Operating temperature -40-70°C



- **Calibration of Sharp IR**

The sensor was adjusted to make detection between 20cm and 30 cm. The 30 cm is the free space between the position of IR sensor and beyond the entranceas .

That was achieved by setting the free distance (20-30 cm) in the IR code, measuring a distance of 30 cm from the IR and comparing the actual measurement with the measurement in the serial monitor of Arduino. So, if there is a car isin a range greater than 20 cm and less than 30 cm in addition to a valid RFID tag, the gate will open. If the car is not within this distance, that means it gets in and the gate will close.

- **Sharp IR Connection**

❖ This type of sensors has three wires:

• Red is VCC the power supply.

• Black is the ground.

• Yellow is the IR analog pulse

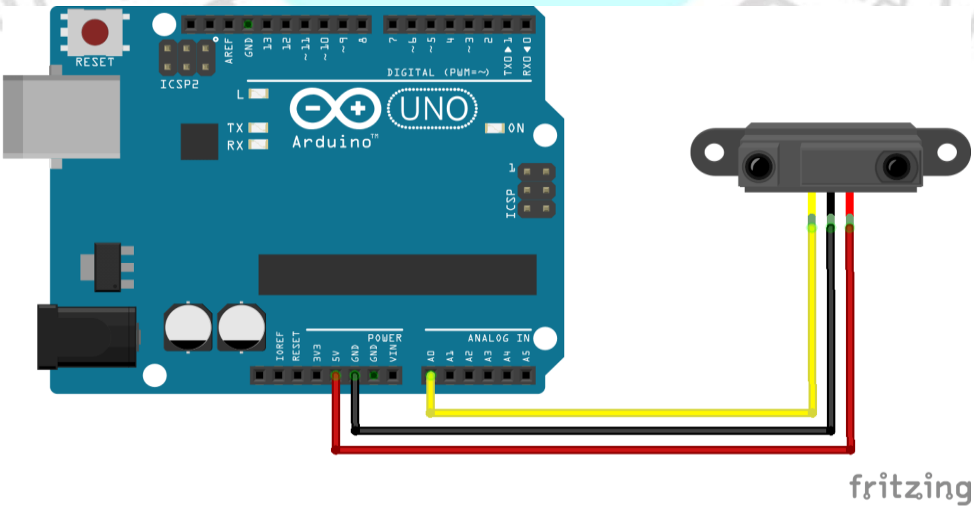
- **IR and Arduino Interface**

❖ Connecting of IR sensor with arduino is indicated in the figure

• Red wire can be connected to +5V in Arduino

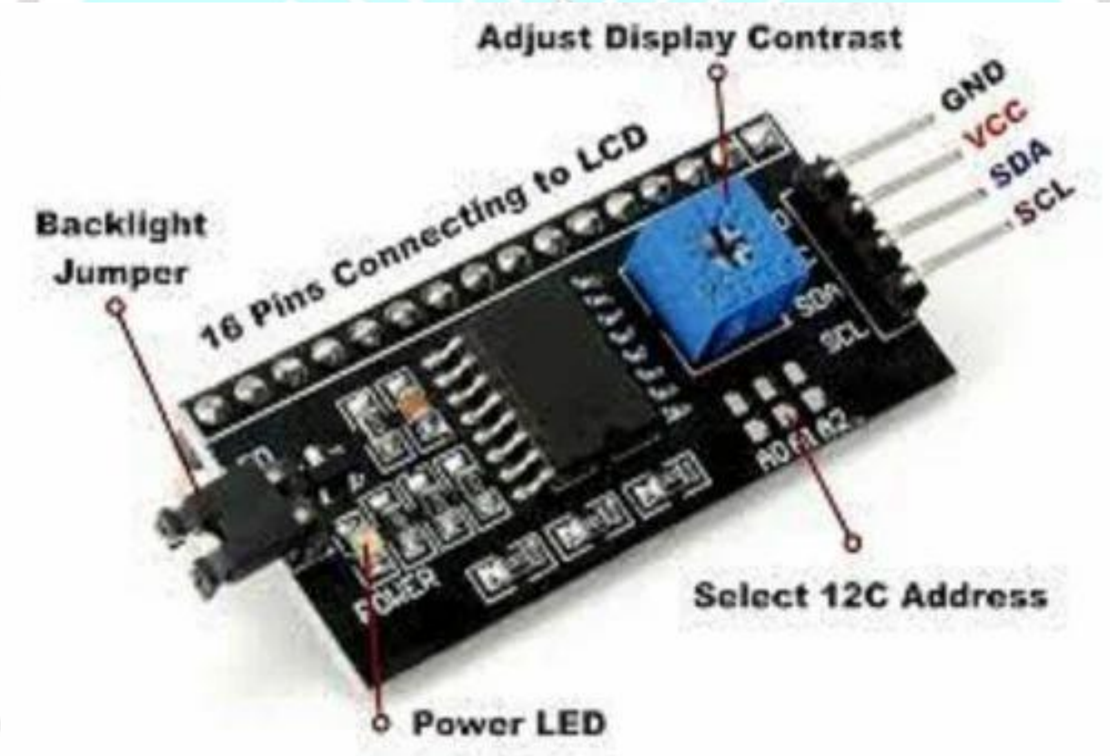
• Black wire is connected to ground in Arduino

• Yellow wire is the analog pulse which can be connected to analog inputs in Arduino



## 5. I2C LCD MODULE :

The most notable feture of these modules is their I2C interface, which simplifies the connection to microcontrollers and other devices. This means you can control the display withjust a couple of wires (SDA and SCL) instead of many individual pins.




I2C lcd modules are typically compact and come in various sizes, including 16*2(16 characters by 2 lines) and 20*4 (20 characters by 4 lines)displays. The compact size makes them suitable for embedded systems and small-scale projects.

I2c lcd modules feature an LED backlight, allowing you to a adjust the brightness or turn it off when needed. the Backlight makes the text and characters on the display visible in various lighting conditions.

## 6. Arduino USB cable and Power adapter

Another way to power up the Arduino is to plug in a battery or wall adapter into the DC jack. The rating should be 9V DC 100-500mA power adapter, with a 2.1mm barrel plug and positive tip. Right next to the USB jack, there is a jumper with 3 pins. If the jumper is on the two pins nearest USB jack that means you are planning to power the Arduino via the USB cable. If it's on the two pins nearer the DC Jack then it means you are planning to power the Arduino using a 9V battery or wall adapter.

# CHAPTER -3

## Working

While making all the connections firstly we have Aurdino in which it has 6 analog inputs and 10 digital input/output pins and its operating voltage is 5 v and input voltage is 20 v and servo motor provides rotatory actuatory of motor for precise control in terms of angular position velocity and acceleration and LCD is used to display the things then we require and potentiometer when the wiper rotates it shows output on LCD screen and push buttons helps to detect available parking spaces facilating the task of drivers looking for vacant parking spots in closed spaces .

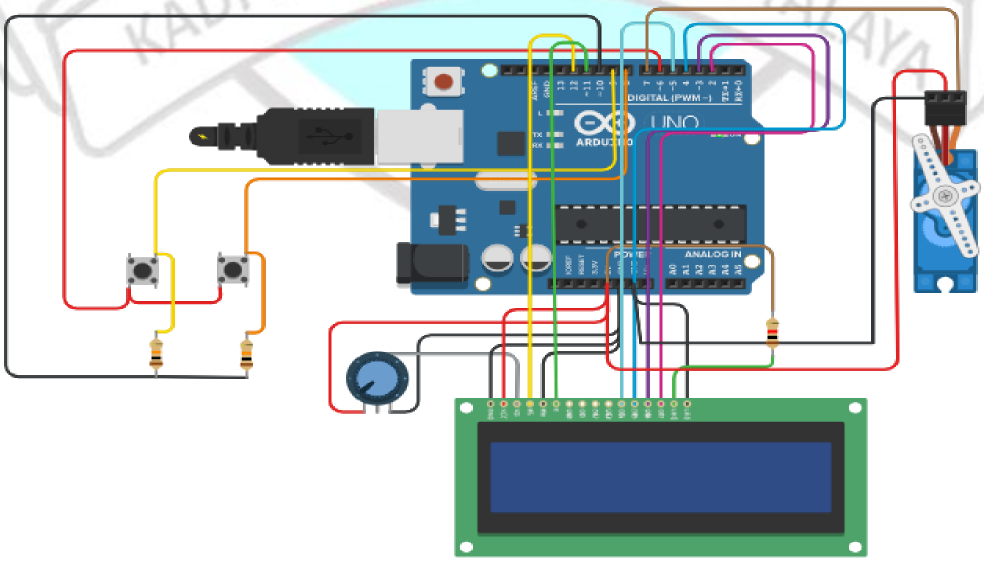
when we push a push button accordingly our servo motor rotates then it shows output on LCD using potentiometer shows “space left of 7 cars” through entry port then accordingly it “shows space left for 6 cars” after then again pressing on push button multiple times then at last it displays “no space left” means all the 7 cars are inside in parking lot there is no space left or car to park so to park more cars inside the parking lot we try to exit cars in parking lot so this is done by exit port when cars exit in parking lot then when we press a push button accordingly servo motor rotates it shows output on LCD “space left for one car” similarly second car exit accordingly servo motor rotates and shows output on LCD “space left for 2 cars” similarly all the cars exit in parking lot and at last it shows “Space left for 7 car”.

Then again same process happens in Entry and Exit push buttons in this way the smart parking system project works.

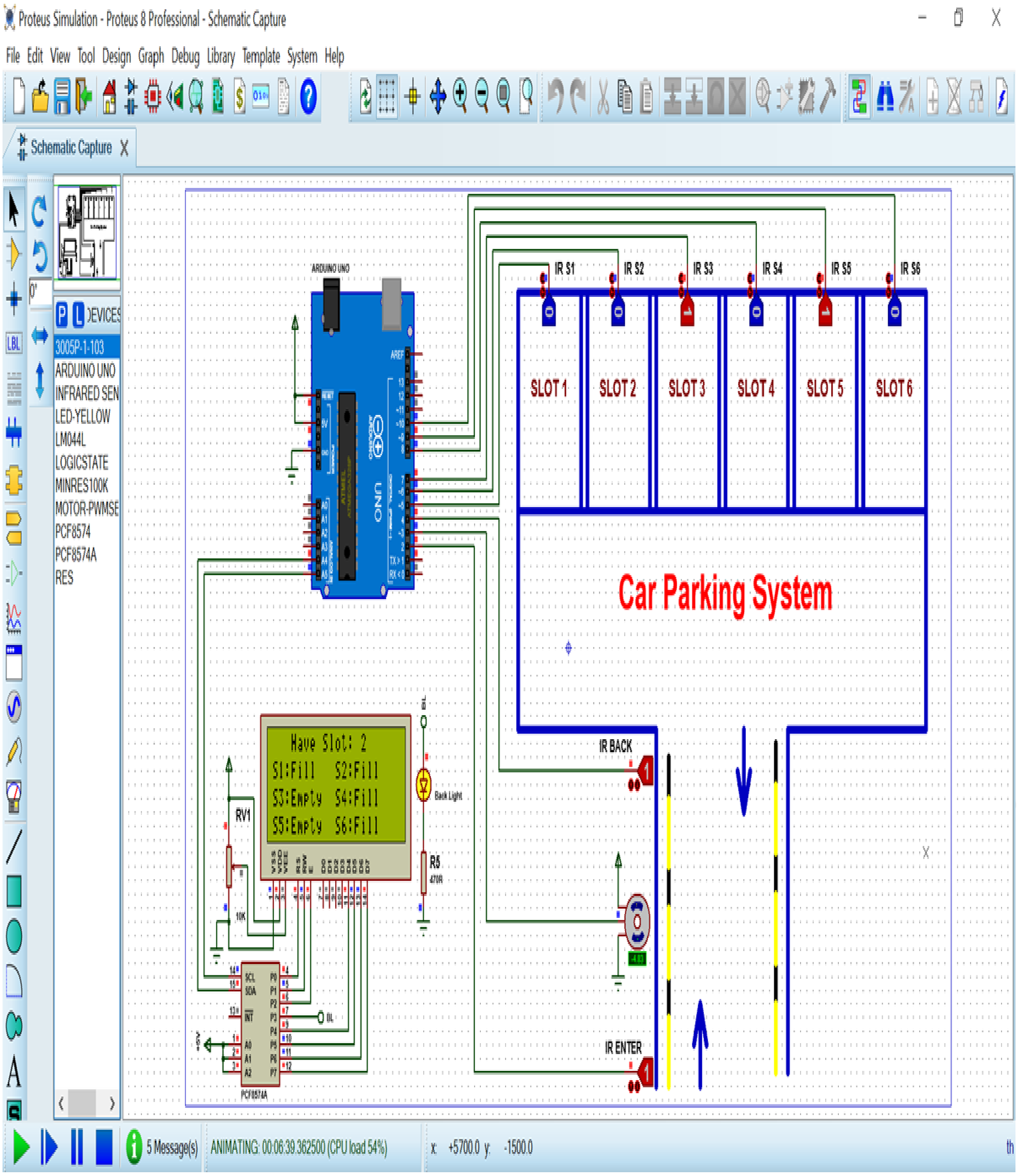
# CHAPTER -4

## Simulations

To simulate our project car parking system using Aurdino first we need a resistor of value 10 kilo ohm each then we require two push buttons at the entry of parking lot and exit port of parking lot out of which entry terminal push button is connected to pin 8 of Aurdino via resistor by pin 10 of Aurdino . Now exit terminal push button is connected to pin 9 of Aurdino connected via entry terminal push button through resistors by pin 6 of Aurdino .Then we require potentiometer to give contrast on lcd output it has three terminals terminal 1 terminals 2 and wiper for rotation and terminal 1 connected to 5 v of aurdino wiper connected to contrast of LCD display and terminal 2 is connected to ground pin of Aurdino . Now we require Servo motor for three pin Ground pin Power pin and signal pin . Now ground pin is connected to ground pin of Aurdino and power pin is connected to 5 v of Aurdino and signal pin is connected to pin 7 of Aurdino . Now we require LCD display in which LCD cathode to ground pin of aurdino and LCD anode is connected to resistor through 5 v of Aurdino and DB7 pin of LCD is connected to pin 2 of Aurdino then DB6 pin is connected to pin 3 of Aurdino and DB5 pin is connected to pin 4 of Aurdino and DB4 pin is connected to pin 5 of Aurdino and enable pin is connected to pin 11 of Aurdino then Read/write pin is connected ground of Aurdino and register pin is connected to pin 12 of Aurdino and power is connected to 5v of Aurdino and ground is connected to ground of Aurdino.



# SCHEMATIC DIAGRAM



# CHAPTER -5

## Code of Arduino Software

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 2, 1, 0, 4, 5, 6, 7, 3, POSITIVE);

Servo myservo;

#define ir_enter 2
#define ir_back  4

#define ir_car1 5
#define ir_car2 6
#define ir_car3 7
#define ir_car4 8
#define ir_car5 9
#define ir_car6 10

int S1=0, S2=0, S3=0, S4=0, S5=0, S6=0;
int flag1=0, flag2=0;
int slot = 6;

void setup(){
Serial.begin(9600);
```

```
pinMode(ir_car1, INPUT);
pinMode(ir_car2, INPUT);
pinMode(ir_car3, INPUT);
pinMode(ir_car4, INPUT);
pinMode(ir_car5, INPUT);
pinMode(ir_car6, INPUT);

pinMode(ir_enter, INPUT);
pinMode(ir_back, INPUT);

myservo.attach(3);
myservo.write(90);

lcd.begin(20, 4);
lcd.setCursor (0,1);
lcd.print("   Car  parking  ");
lcd.setCursor (0,2);
lcd.print("     System     ");
delay (2000);
lcd.clear();

Read_Sensor();

int total = S1+S2+S3+S4+S5+S6;
slot = slot-total;
}
```

```
void loop(){

Read_Sensor();

lcd.setCursor (0,0);
lcd.print("   Have Slot: ");
lcd.print(slot);
lcd.print("    ");

lcd.setCursor (0,1);
if(S1==1){lcd.print("S1:Fill ");}
    else{lcd.print("S1:Empty");}

lcd.setCursor (10,1);
if(S2==1){lcd.print("S2:Fill ");}
    else{lcd.print("S2:Empty");}

lcd.setCursor (0,2);
if(S3==1){lcd.print("S3:Fill ");}
    else{lcd.print("S3:Empty");}

lcd.setCursor (10,2);
if(S4==1){lcd.print("S4:Fill ");}
    else{lcd.print("S4:Empty");}
```

```
 lcd.setCursor (0,3);
if(S5==1){lcd.print("S5:Fill ");}
    else{lcd.print("S5:Empty");}

lcd.setCursor (10,3);
if(S6==1){lcd.print("S6:Fill ");}
    else{lcd.print("S6:Empty");}

if(digitalRead (ir_enter) == 0 && flag1==0){
if(slot>0){flag1=1;
if(flag2==0){myservo.write(180); slot = slot-1;}
}else{
lcd.setCursor (0,0);
lcd.print(" Sorry Parking Full ");
delay(1500);
}
}

if(digitalRead (ir_back) == 0 && flag2==0){flag2=1;
if(flag1==0){myservo.write(180); slot = slot+1;}
}

if(flag1==1 && flag2==1){
delay (1000);
myservo.write(90);
flag1=0, flag2=0;
```

```
}

delay(1);
}

void Read_Sensor(){
S1=0, S2=0, S3=0, S4=0, S5=0, S6=0;

if(digitalRead(ir_car1) == 0){S1=1;}
if(digitalRead(ir_car2) == 0){S2=1;}
if(digitalRead(ir_car3) == 0){S3=1;}
if(digitalRead(ir_car4) == 0){S4=1;}
if(digitalRead(ir_car5) == 0){S5=1;}
if(digitalRead(ir_car6) == 0){S6=1;}
}
```

# CHAPTER -6

## Further Modifications

### Automatic Ticketing :

An automatic ticket generating system can be added to this project. The parts of to be added for implementing this extension are as follows:

➢ Object detection: The object detection model will analyze the size of the vehicle and detect the type of it. And accordingly it will provide a suitable parking slot for your vehicle.

➢ Optical Character Recognition: For an automatic ticketing system, it needs to have a registration number and brand of the vehicle on the parking ticket. So for extracting the registration number and the brand, we will use the OCR model to detect the number plate and the text on it. To generate tickets.

**Mobile App for Realtime Data:** We can also have the mobile app or WebApp to display the real time status of parking lots in the parking area in nearby locations. The main components of the Mobile app can be:

➢ List of Parking Areas in the location: Users can see the list of all the parking areas available in the nearby locations. Location can be automatically detected by the app ( depends on the user).

➢ The Parking Slot Map: Users can also see the map of the parking slots in any particular parking area. This will help the user to quickly find the Parking slot he is allotted.

**Extension To The Mobile App:** Users can see the real time status of the total number of slots and free slots in the nearby Parking Areas. But they still have to go and get a token manually for parking the car.With this extension, users will be able to book the parking slot in advance from the application itself. They do not have to worry about whether there will be any spot available for them at 3 pm tomorrow or not.

They can have tickets in advance.

# CONCLUSION

This study has proposed a parking system that improves performance by reducing the number of users that fail to find a parking space and minimizes the costs of moving to the parking space. Our proposed architecture and system has been successfully simulated and implemented in a real situation. The results show that our algorithm significantly reduces the average waiting time of users for parking. Our results closely agree with those of our proposed mathematical models. The simulation of our system achieved the optimal solution when most of the vehicles successfully found a free parking space. The average waiting time of each car park for service becomes minimal, and the total time of each vehicle in each car park is reduced. In our future study, we will consider the security aspects of our system as well as implement our proposed system in large scales in the real world.

If this smart parking system with all the mentioned modifications is implemented, then surely this would be a revolution in the field of parking. This system would save time, fuel, labour and thus would decrease the cost. All we need is just a single time investment. We would be able to check the availability of parking space a well as book a slot just by a single click. Since the details of the vehicle as well as the driver are being collected, so they can be referred for any discrepancy in the parking slots, thus improving the security of our vehicles too.

# REFERENCES


1. https://en.wikipedia.org/wiki/Servomotor
2. https://whatis.techtarget.com/definition/LCD-liquid-crystal-display.
3. https://robu.in/ir-sensor-working/
4. https://www.quisure.com/blog/faq/what-is-the-principle-of-the-push-button-switches